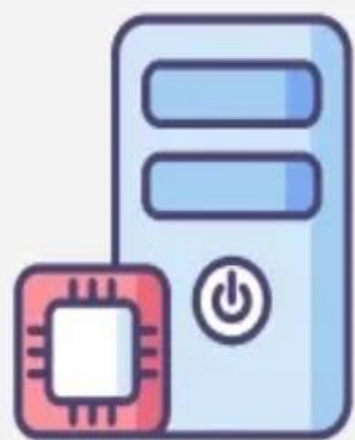


Hypervisor

A hypervisor (or Virtual Machine Monitor, VMM) is software that lets multiple operating systems run on a single physical machine. It manages hardware resources (CPU, memory, storage) and allocates them to virtual machines (VMs) without interference. This improves hardware utilization, reduces costs, and provides flexibility in cloud and server environments.

How It Works:

A hypervisor runs on hardware or a host OS to create and manage virtual machines (VMs), each with its own virtual CPU, memory, storage, and network. It intercepts guest OS requests and translates them to physical hardware, ensuring isolation, security, and stability.



Hardware



Hypervisor



OS



- TYPES:

There are two types:

1. Type 1 Hypervisor

- A Type 1 hypervisor runs directly on the host's hardware. It doesn't rely on a host operating system. This architecture offers better performance and security because there is no intermediary OS. It's the standard for enterprise-level data centers and cloud providers like Amazon Web Services (AWS) and Microsoft Azure.
- **Type-1 Hypervisor (Bare-Metal Hypervisor) – Full Points**
- A **Type-1 hypervisor** is installed **directly on the physical hardware** of the system.
- It **does not require a host operating system**, making it more efficient.
- It directly **controls and manages hardware resources** such as CPU, memory, storage, and network.
- Provides **high performance** because there is no extra OS layer.
- Offers **better security** due to minimal attack surface.
- Ensures strong **isolation between virtual machines**.
- Highly **reliable and stable** for continuous operation.
- Supports **multiple virtual machines**, each running its own OS.
- Mainly used in **cloud computing platforms and enterprise data centers**.
- Suitable for **mission-critical applications**.

- **Examples:** VMware ESXi, Microsoft Hyper-V, KVM (Kernel-based Virtual Machine), and Xen.
- **Pros:**
 - High performance (direct hardware access).
 - Strong security (no intermediate OS layer).
 - Suitable for mission-critical workloads.
- **Cons:**
 - Requires dedicated hardware.
 - Setup and management are complex compared to Type-2.

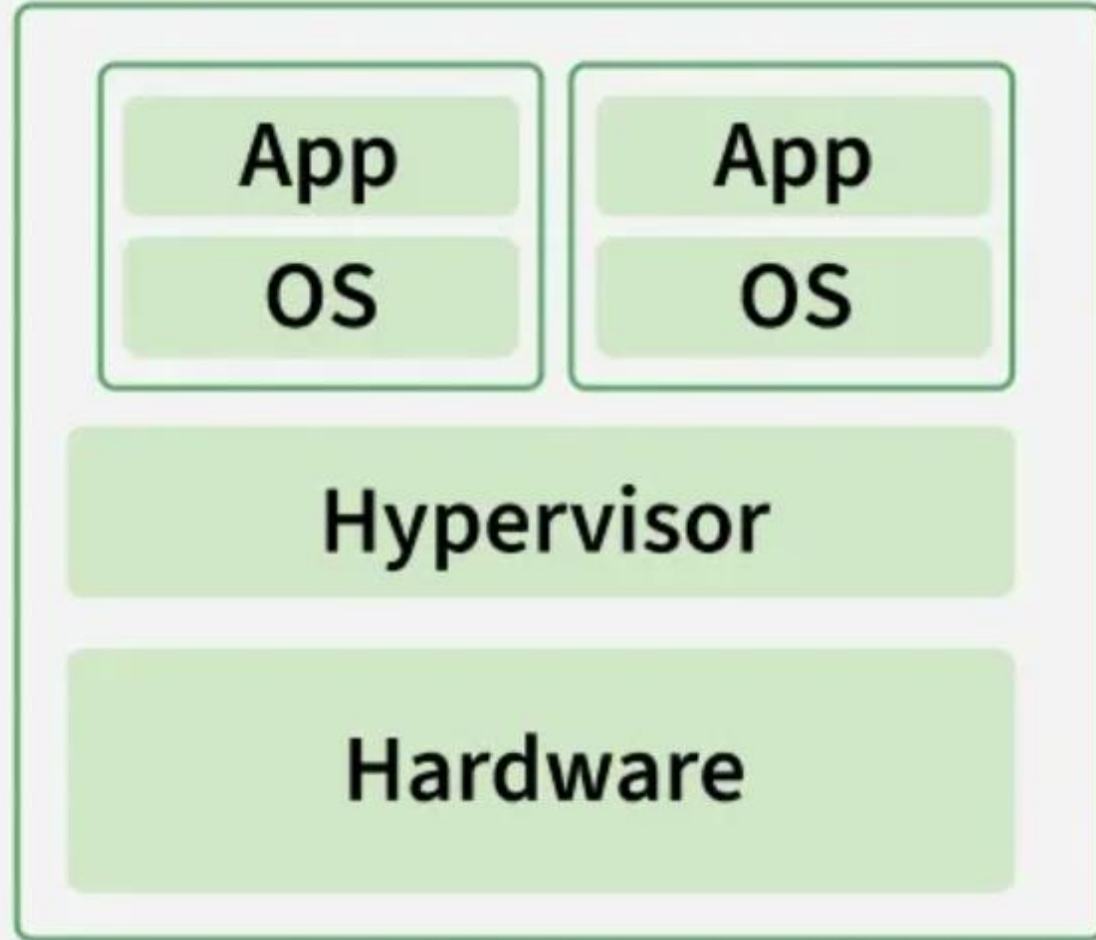
- **2. Type 2 Hypervisor**

- A **Type 2 hypervisor** runs on top of a conventional operating system (like Windows, macOS, or Linux). It's essentially an application within the host OS. This type is generally used for desktop virtualization, development, and testing environments where a user needs to run multiple OSs on their personal computer. Performance is slightly lower than Type 1 due to the overhead of the host OS.

- A **Type-2 hypervisor** is installed on top of an existing host operating system.
- It depends on the **host OS** to access hardware resources.
- Hardware management is done **indirectly** through the host OS.
- Performance is **lower than Type-1 hypervisor** due to the extra OS layer.
- Easier to **install, configure, and use**.
- Suitable for **personal use, learning, testing, and development**.
- Provides good **isolation** between virtual machines, but less secure than Type-1.
- Supports running **multiple operating systems** on a single machine.
- Not preferred for **enterprise or production environments**.



- **Examples:** Oracle VM VirtualBox, VMware Workstation, and Parallels Desktop.
- **Pros:**
 - Easy to install and use.
 - Useful for development, testing, and malware analysis.
 - Provides good host–guest integration features.
- **Cons:**
 - Slower performance (no direct hardware access).
 - Security depends on the host OS; compromise of host may affect guests.



**Type 1 (Bare Metal)
Virtualization**



**Type 2 (Hosted)
Virtualization**

- **Importance of Hypervisors**

- **1. Efficient Resource Utilization**

Hypervisors allow multiple virtual machines (VMs) to run on a single physical server. This ensures maximum use of CPU, RAM, storage, and network resources, reducing hardware wastage.

- **2. Cost Reduction**

Organizations save money by reducing the number of physical servers required. It lowers hardware costs, electricity consumption, cooling expenses, and maintenance costs.

- **3. Isolation and Security**

Each virtual machine operates independently. If one VM fails or is attacked by malware, other VMs remain unaffected, improving overall system security.

- **4. Scalability**

- Hypervisors allow easy allocation or removal of resources like memory and CPU power. This makes it simple to scale applications based on demand.

- **5. Flexibility**

Different operating systems (such as Windows, Linux, etc.) can run simultaneously on the same physical hardware, providing greater flexibility.